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Assignment 3

Small Object Detection Using YOLO

The goal of this report is to demonstrate training a model for small object detection specifically detection of traffic lights and signs from images. For model training a fraction of the Mapillary traffic sign dataset were used which were then randomly split into training (13102 images), validation (1746 images) and testing (2629 images) sets. The Mapillary traffic Sign Dataset is a diverse imagery of dataset with bounding box annotation for detecting and classifying traffic signs around the world. It includes over 300 traffic sign classes. The dataset annotations were first converted from the MTSD format which uses a bounding box around the object of interest into YOLO format which requires a class ID followed by horizontal and vertical center, width and height of the target object in the image. The validation set of the data is uploaded to Github repository for reference.

Initial inference calls were made using yolov8n.pt model which is a real-time object detection pre-trained model trained to recognize 80 different classes of the objects. Eight sample predictions on the testing data are shown in figure 1, where most of the car predictions are correct, however some of the predictions are ambiguous such as the cow prediction on the seventh image. Find the initial prediction on the rest of the test dataset in pre_trained_prediction.json file.

Figure 1. Initial prediction on testing image using yolov8 moel

The model was further trained on the mentioned dataset of small traffic objects with the following training configuration (epochs = 20, imgsz = 1280, batch = 8, patience = 10), and the best weights were saved from the during training. Using the best weights model another set of

predictions were made.The model prediction focus significantly shifts from detecting object of various types (eg. cars, animals) to only detect road signs as shown in Figure 2.

Figure 2. Pre-trained and fine-tunned model prediction on the small objects.

A new model was trained with the same base model (YOLOv8n) and resolution (1280px) as the above but added aggressive data augmentation — rotation ±15° , scale variation ±90% , shear ±5° , translation ±20%, vertical flip at 30%, image blending at 15% (mixup=0.15), extended mosaic cutoff, boosted color jitter (hsv_h=0.02, hsv_s=0.8, hsv_v=0.5), and random erasing (erasing=0.3).
However the aggressively augmented model underperformed the compared to the previous model y metric — mAP50 dropped from 0.667 to 0.638, mAP50-95 fell from 0.449 to 0.389, and recall declined from 0.594 to 0.562, while precision stayed roughly flat (0.768 vs 0.758). The heavy transformations effectively made each training sample harder to learn from, and 20

epochs was not enough for the model to converge under this more challenging distribution. Comparison of prediction shown in Figure 3.

Figure 3. Baseline trained model on the new data vs aggressively augmented model